

# Sheaves & Proofs

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## Desired background

Some exposure to proof theory (sequent calculus)  
and semantics (structures, satisfaction, soundness, completeness)  
for classical predicate logic.

## Not required background

- Intuitionistic logic (sequent calculus, Kripke models)
- Basic category theory (categories, functors, natural transformations)
- Basic point-set topology (topologies, covers)

## Part 1

# Coverage Semantics over Partial Orders

(The logic of localic toposes)

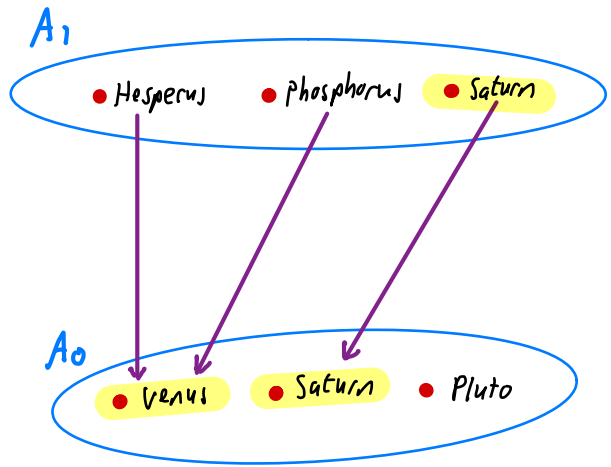
- Kripke semantics
- Coverage semantics
- Cut-elimination via  
constructive completeness
- The **77**-dense coverage.

# (Inverted) Kripke semantics

A Kripke Model is given by

- A partial order  $(W, \leq)$  of Worlds
- To every  $w \in W$  a set  $A_w$
- Transition functions  $(t_{vw} : A_w \rightarrow A_v)_{v \leq w}$   
Satisfying  $t_{ww} = id_{A_w}$ ;  $u \leq v \leq w \Rightarrow t_{uw} = t_{uv} \circ t_{vw}$
- For every predicate  $P$  of arity  $k$   
Subsets  $P_w \subseteq (A_w)^k$  satisfying  
 $v \leq w \Rightarrow t_{vw}(P_w) \subseteq P_v$ .

$$W = \begin{array}{c} 1 \\ | \\ 0 \end{array}$$



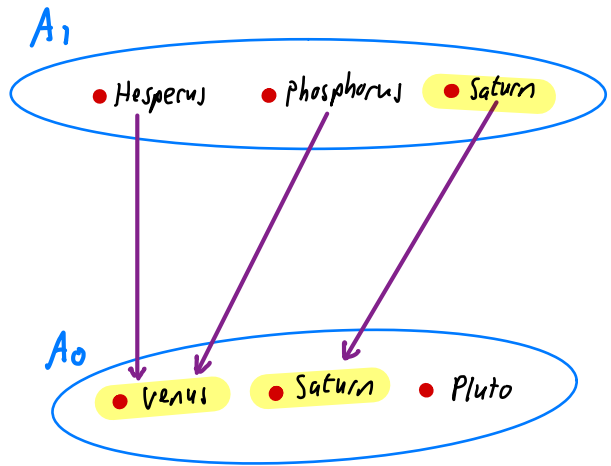
 The planet predicate

# Category-theoretic Formulation

A Kripke model is given by

- A partial order  $(W, \leq)$  of Worlds
- A functor  $A: W^{op} \rightarrow \underline{\text{Set}}$   
(i.e., a presheaf  $A \in \text{Psh}(W)$ )
- For every predicate  $P$  of arity  $k$   
a subpresheaf  $P \subseteq A^k$

$$W = \begin{array}{c} \bullet \\ \vdots \\ \bullet \\ \vdots \\ \bullet \end{array} \begin{array}{c} 1 \\ \\ 0 \\ \\ \end{array}$$



 The planet predicate

Notation

|                                              |                  |
|----------------------------------------------|------------------|
| $\text{Venus} = \text{top}(\text{Hesperus})$ | (previous slide) |
| $= A(0<1)(\text{Hesperus})$                  | (this slide)     |
| $= \text{Hesperus} \cdot 0$                  | (henceforth)     |

# Kripke forcing relation

$$w \Vdash \phi$$

$$w \Vdash P(a_1, \dots, a_k) \Leftrightarrow (a_1, \dots, a_k) \in P_w$$

$$w \Vdash a_1 = a_2 \Leftrightarrow a_1 = a_2$$

$$w \Vdash \phi \wedge \psi \Leftrightarrow w \Vdash \phi \text{ and } w \Vdash \psi$$

$$w \Vdash \phi \vee \psi \Leftrightarrow w \Vdash \phi \text{ or } w \Vdash \psi$$

$$w \Vdash \perp \Leftrightarrow \text{false}$$

$$w \Vdash \exists x \phi \Leftrightarrow \text{there exists } a \in A_w \text{ s.t. } w \Vdash \phi[x:=a]$$

$$w \Vdash \phi \rightarrow \psi \Leftrightarrow \text{for all } v \leq w \quad v \Vdash \phi \cdot v \text{ implies } v \Vdash \psi \cdot v$$

$$w \Vdash \forall x \phi \Leftrightarrow \text{for all } v \leq w \text{ and } a \in A_v \quad v \Vdash (\phi \cdot v)[x:=a]$$

$1 \Vdash \text{Hesperus} = \text{Phosphorus}$

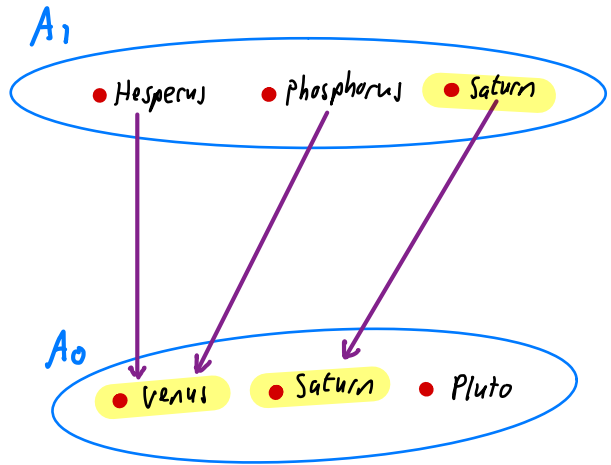
$0 \Vdash \text{Hesperus} \cdot 0 = \text{Phosphorus} \cdot 0$

$1 \Vdash \neg (\text{Hesperus} = \text{Phosphorus})$

$1 \Vdash H=P \vee \neg (H=P)$

$1 \Vdash \neg \neg (H=P)$

$$W = \begin{matrix} & 1 \\ & \vdots \\ & 0 \end{matrix}$$



 The planet predicate

$\neg \phi := \phi \rightarrow \perp$

$w \Vdash \neg \phi \Leftrightarrow \forall v \leq w. v \nVdash \phi$

$w \Vdash \neg \neg \phi \Leftrightarrow \forall v \leq w. \neg \neg \exists u \leq v. u \Vdash \phi$

## Meta theorems for Kripke semantics

Monotonicity If  $w \Vdash \phi$  and  $v \leq w$  then  $v \Vdash \phi$ .

Soundness If  $\phi$  is provable in intuitionistic predicate logic then for all Kripke models  $(W, \dots)$  and  $w \in W$ ,  $w \Vdash \phi$ .

Completeness If, for all Kripke models  $(W, \dots)$  and  $w \in W$ ,  $w \Vdash \phi$  then  $\phi$  is provable in intuitionistic predicate logic.

Completeness is not provable in a constructive meta-theory.

To achieve constructive completeness, we need more general notion of model.



# Coverage base

a.u.a. base for a  
Grothendieck topology  
on a poset

A coverage base on a poset  $\mathcal{W}$  is a relation  $\triangleright$  relating pairs  $C \triangleright w$ , where  $w \in \mathcal{W}$  and  $C \subseteq \downarrow w$ , satisfying:

$$\downarrow w := \{v \in \mathcal{W} \mid v \leq w\}$$

$$\triangleright \subseteq \sum_{w \in \mathcal{W}} \mathcal{P}(\downarrow w)$$

Reflexivity  $\{w\} \triangleright w$

Transitivity If  $C \triangleright w$  and, for all  $v \in C$ ,  $C_v \triangleright v$  then  $\bigcup_{v \in C} C_v \triangleright w$ .

Stability If  $C \triangleright w$  and  $w' \leq w$  then there exists  $D \triangleright w'$  such that, for all  $v' \in D$  there exists  $v \in C$  such that  $v' \leq v$ .

## Example coverage bases

① The identity coverage  $\{w\} \triangleright w$  on any poset  $W$

② For any topological space  $T$ , define

$$W = \mathcal{O}(T) \quad (\text{open sets})$$

$$v \leq w \iff v \subseteq w$$

$$C \triangleright w \iff \bigcup C = w \quad (C \text{ covers } w)$$

Exercise Prove that ② satisfies properties of a cover base!

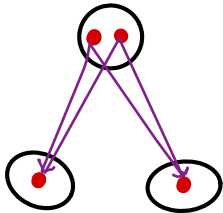
A separated Model is given by

- A partial order  $(W, \leq)$  of worlds with coverage base  $\triangleright$
- To every  $w \in W$  a set  $A_w$
- Transition functions  $(t_{vw} : A_w \rightarrow A_v)_{v \leq w}$  s.t.  $u \leq v \leq w \Rightarrow t_{uw} = t_{uv} \circ t_{vw}$
- $C \triangleright w \Rightarrow \forall a, b \in A_w (\forall v \in C, t_{vw}(a) = t_{vw}(b)) \Rightarrow a = b$  . ( $\triangleleft$ -separated)
- For every predicate  $P$  of arity  $k$  subsets  $P_w \subseteq (A_w)^k$  satisfying
  - $\forall \vec{a} \in A_w^k (\vec{a} \in P_w \Rightarrow \forall v \leq w, t_{vw}(\vec{a}) \in P_v)$
  - $C \triangleright w \Rightarrow \forall \vec{a} \in A_w^k (\forall v \in C, t_{vw}(\vec{a}) \in P_v) \Rightarrow \vec{a} \in P_w$  ( $P$  is  $\triangleleft$ -closed)

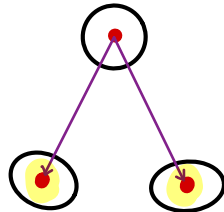
# Category-theoretic Formulation

A separated model is given by

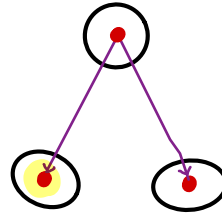
- A partial order  $(W, \leq)$  of worlds
- A  $\triangleright$ -separated presheaf  $A : W^{op} \rightarrow \underline{\text{Set}}$
- For every predicate  $P$  of arity  $k$ , a  $\triangleright$ -closed subpresheaf  $P \subseteq A^k$



not separated



not closed



a separated model

$$W = \bigwedge_{0 \ 0'}^1$$

$$\{0, 0'\} \triangleright 1$$

$$\{0\} \not\triangleright 1$$

$$\{0'\} \not\triangleright 1$$

## New forcing clauses

$w \Vdash \phi \vee \psi \iff$  there exists a cover  $C \triangleright w$  such that,  
for every  $v \in C$ ,  $v \Vdash \phi \cdot v$  or  $v \Vdash \psi \cdot v$

$w \Vdash \perp \iff \emptyset \triangleright w$

$w \Vdash \exists x \phi \iff$  there exists a cover  $C \triangleright w$  such that,  
for every  $v \in C$ ,  
there exists  $a \in \underline{A}_v$  s.t.  $v \Vdash (\phi \cdot v)[x := a]$

N.B.

- The clauses for predicates,  $=$ ,  $\wedge$ ,  $\rightarrow$ ,  $\forall$  are as for Kripke semantics

For our example coverage bases

(1) The identity coverage base  $\{w\} \triangleright w$  on any poset  $W$

Recovers Kripke semantics

(2) The topological cover relation for  $W = \mathcal{O}(T)$ .

Recovers topological semantics of intuitionistic logic

## Meta theorems for coverage semantics

Monotonicity If  $w \Vdash \phi$  and  $v \leq w$  then  $v \Vdash \phi$ .

Cover property If  $C \supseteq w$  satisfies, for all  $v \in C$ ,  $v \Vdash \phi$ ,  
then  $w \Vdash \phi$ . ] See warning at the end of Part 2

Soundness If  $\phi$  is provable in intuitionistic predicate logic  
then for all separated models  $(W, \dots)$  and  $w \in W$ ,  $w \Vdash \phi$ .

Completeness If, for all separated models  $(W, \dots)$  and  $w \in W$ ,  $w \Vdash \phi$   
then  $\phi$  is provable in intuitionistic predicate logic

Completeness is now provable in a constructive meta-theory!

## Intuitionistic sequent Calculus

Predicate symbols  $P(x_1, \dots, x_n)$  ( $n \geq 0$ )

No function symbols (in particular no constants) (for brevity of exposition)

Equality  $t_1 = t_2$

[The only terms are variables, but we use  $t, t_1, \dots$  as meta-variables in places where arbitrary terms would be allowed in presence of function symbols.]

Sequent:  $\Theta; \Gamma \Rightarrow \psi$

where  $\Theta$  is finite set of variables

$\Gamma$  is finite set of  $\Theta$ -formulas,  $\psi$  is  $\Theta$ -formula

A  $\Theta$ -formula is a formula whose free variables are contained in  $\Theta$ .



# Rules

$$\frac{}{\Theta ; \Gamma, P(\vec{t}) \Rightarrow P(\vec{t})}$$

$$\frac{\Theta ; \Gamma, \phi, \phi' \Rightarrow \psi}{\Theta ; \Gamma \ni \phi \wedge \phi' \Rightarrow \psi}$$

$$\Theta ; \Gamma \ni \phi \wedge \phi' \Rightarrow \psi$$

$$\frac{\Theta ; \Gamma \Rightarrow \psi \quad \Theta ; \Gamma \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \wedge \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \wedge \psi'$$

$$\frac{\Theta ; \Gamma, \phi \Rightarrow \psi \quad \Theta ; \Gamma, \phi' \Rightarrow \psi}{\Theta ; \Gamma \ni \phi \vee \phi' \Rightarrow \psi}$$

$$\Theta ; \Gamma \ni \phi \vee \phi' \Rightarrow \psi$$

$$\frac{\Theta ; \Gamma \Rightarrow \psi}{\Theta ; \Gamma \Rightarrow \psi \vee \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \vee \psi'$$

$$\frac{\Theta ; \Gamma \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \vee \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \vee \psi'$$

$$\frac{}{\Theta ; \Gamma \ni \perp \Rightarrow \psi}$$

$$\frac{\Theta ; \Gamma, \phi' \Rightarrow \psi \quad \Theta ; \Gamma \Rightarrow \phi}{\Theta ; \Gamma \ni \phi \rightarrow \phi' \Rightarrow \psi}$$

$$\Theta ; \Gamma \ni \phi \rightarrow \phi' \Rightarrow \psi$$

$$\frac{\Theta ; \Gamma, \psi \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \rightarrow \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \rightarrow \psi'$$

$$\frac{\Theta \wp x; \Gamma, \phi \Rightarrow \psi}{\Theta; \Gamma \ni \exists x \phi \Rightarrow \psi}$$

$$\frac{\Theta; \Gamma' \Rightarrow \psi[t/x]}{\Theta; \Gamma \Rightarrow \exists x \psi}$$

$$\frac{\Theta; \Gamma', \phi[t/x] \Rightarrow \psi}{\Theta; \Gamma \ni \forall x t \Rightarrow \psi}$$

$$\frac{\Theta \wp x; \Gamma \Rightarrow \psi}{\Theta; \Gamma \Rightarrow \forall x \psi}$$

$$\frac{\Theta; \Gamma \Rightarrow \psi[s/z]}{\Theta; \Gamma \ni s=t \Rightarrow \psi[t/z]}$$

$$\frac{\Theta; \Gamma \Rightarrow \psi[t/z]}{\Theta; \Gamma \ni s=t \Rightarrow \psi[s/z]}$$

$$\frac{}{\Theta; \Gamma \Rightarrow t=t}$$

$$\frac{\Theta; \Gamma \Rightarrow \phi \quad \Theta; \Gamma, \phi \Rightarrow \psi}{\Theta; \Gamma \Rightarrow \psi} \text{ cut}$$

Define  $\Theta; \Gamma \vdash \psi$   $:\Leftrightarrow$  the sequent  $\Theta; \Gamma \Rightarrow \psi$  is provable

$\Theta; \Gamma \vdash_{cf} \psi$   $:\Leftrightarrow$  the sequent  $\Theta; \Gamma \Rightarrow \psi$  is cut-free provable.

Lemma (admissible rules with/without cut)

- (identity)  $\Theta; \Gamma, \psi \vdash_{cf} \psi$
- (weakening)  $\Theta; \Gamma \vdash \psi \Rightarrow \Theta, \theta'; \Gamma, \Gamma' \vdash \psi$   
 $\Theta; \Gamma \vdash_{cf} \psi \Rightarrow \Theta, \theta'; \Gamma, \Gamma' \vdash_{cf} \psi$

Proofs Straightforward inductions.

Theorem (Cut admissibility)  $\Theta; \Gamma \vdash_{cf} \phi$  and  $\Theta; \Gamma, \phi \vdash_{cf} \psi \Rightarrow \Theta; \Gamma \vdash_{cf} \psi$

We shall give a semantic proof, by establishing

Soundness  $\Theta; \Gamma \vdash \psi \Rightarrow \Theta; \Gamma \models \psi$

Cut-free completeness  $\Theta; \Gamma \models \psi \Rightarrow \Theta; \Gamma \vdash_{cf} \psi$

where  $\models$  is semantic entailment w.r.t. separated models.

The proof will be carried out in a constructive meta-theory, so (in principle) it's possible to extract from the proof an algorithm that maps derivations with cuts to cut-free derivations.

Related work: Beth, Veldman et.al., Palmgren, Avigad

# Semantic entailment

$$\Theta; \Gamma \models \psi$$

$:\Leftrightarrow$

for every poset  $W$  and coverage base  $\triangleright$  on  $W$ ,

for every separated model  $(A: W^{op} \rightarrow \underline{\text{set}}, \{p_w\}_w)$ ,

for every  $w \in W$ ,

for every  $\rho: \Theta \rightarrow A_w$

if, for every  $\phi \in \Gamma$ ,  $w \Vdash \phi[\rho]$ ,

then  $w \Vdash \psi[\rho]$

$\forall w$

$\Theta; \Gamma \models_m \psi$

Proof of soundness:  $\Theta; \Gamma \vdash \psi \Rightarrow \Theta; \Gamma \models \psi$

Consider any  $m$ , (i.e.,  $\mathbb{W}, \triangleright$  and separated model  $A, \dots$ )

We prove by ind. on derivation of  $\Theta; \Gamma \vdash \psi$  that  $\Theta; \Gamma \models_m \psi$  holds

Exercise!

Proof of cut-free completeness:  $\Theta; \Gamma \models \psi \Rightarrow \Theta; \Gamma \vdash_{cf} \psi$

In Part 1, we consider just the case of intuitionistic propositional logic

- All predicate symbols have arity 0.  
(Predicates of arity 0 are atomic propositions.)
- No variables, no quantifiers  $\forall, \exists$
- No equality  $=$

We build a specific separated model  $\mathcal{M}_{ctx}$

# The modal $M_{ctx}$ (propositional logic)

$W_{ctx} := \{ \Gamma \mid \Gamma \text{ a finite set of formulas} \}$  (The set of contexts)

$\Gamma' \leq \Gamma \iff \Gamma' \supseteq \Gamma$

$C \triangleright_{ctx} \Gamma \iff \frac{C}{\Gamma} \text{ is a derivable context rule}$

$A_{\Gamma} := \emptyset$

$() \in P_{\Gamma} \iff \Gamma \vdash_{cf} P \quad . \quad (\Gamma \Vdash P \iff \Gamma \vdash_{cf} P)$

○ Corrected from lecture. (Thanks to Dominik Wehr)



# Context rules

$$\frac{\Gamma, \phi, \phi'}{\Gamma \ni \phi \wedge \phi'}$$

$$\frac{\Gamma, \phi \quad \Gamma, \phi'}{\Gamma \ni \phi \vee \phi'}$$

$$\frac{}{\Gamma \ni \perp}$$

$$\frac{\Gamma, \phi'}{\Gamma \ni \phi \rightarrow \phi'} \quad \Gamma \vdash_{cf} \phi$$

For any context  $\Gamma$  and set of contexts  $C$ ,

$\frac{C}{\Gamma}$  is a derivable context rule if there exists a derivation tree of contexts with conclusion  $\Gamma$ , in which every leaf is in  $C$ .

Lemma Suppose  $\frac{C}{\Gamma}$  is a derivable context rule.

- (weakening) For any  $\Gamma''$ ,

$$\frac{\{\Gamma', \Gamma'' \mid \Gamma' \in C\}}{\Gamma, \Gamma''}$$

is a derivable context rule

The stability  
of  $\triangleright_{\text{ctx}}$   
follows.

- (sequencing) For any  $\psi$ ,

$$\frac{\{\Gamma' \Rightarrow \psi \mid \Gamma' \in C\}}{\Gamma \Rightarrow \psi}$$

is a cut-free derivable sequent rule.

The  $\Delta$ -closed  
property of  
 $M_{\text{ctx}}$   
follows

Main lemma  $\mathcal{M}_{ctx}$  enjoys the following.

For any  $\Gamma \in \mathcal{W}_{ctx}$  and formula  $\phi$

(1) if  $\phi \in \Gamma$  then  $\Gamma \Vdash \phi$ .

(2) if  $\Gamma \Vdash \phi$  then  $\Gamma \vdash_{cf} \phi$ .

Proof of cut-free completeness:  $\Gamma \models \psi \Rightarrow \Gamma \vdash_{cf} \psi$

Suppose  $\Gamma \models \psi$ .

In particular  $\Gamma \models_{\mathcal{M}_{ctx}} \psi$ .  $\otimes$

By ① of main lemma, for every  $\phi \in \Gamma$ ,  $\Gamma \Vdash \phi$ .

By  $\otimes$ ,  $\Gamma \Vdash \psi$ .

By ② of main lemma,  $\Gamma \vdash_{cf} \psi$ .

□

Main lemma  $M_{ctx}$  enjoys the following.

For any  $\Gamma \in W_{ctx}$  and formula  $\phi$

(1) if  $\phi \in \Gamma$  then  $\Gamma \Vdash \phi$ .

(2) if  $\Gamma \Vdash \phi$  then  $\Gamma \vdash_{cf} \phi$ .

Proof

(1) and (2) are proved simultaneously by induction on  $\phi$ .

□

Case  $\phi \equiv \phi_1 \rightarrow \phi_2$

(1)  $\phi_1 \rightarrow \phi_2 \in \Gamma \Rightarrow \Gamma \Vdash \phi_1 \rightarrow \phi_2$   
-----

Suppose  $\Gamma' \supseteq \Gamma$  and  $\Gamma' \Vdash \phi$ ,

By i.h. (2)  $\Gamma' \vdash_{cf} \phi_1$

By  $\bigcirc$ ,  $\{\Gamma', \phi_2\} \supseteq_{ctx} \Gamma'$

By i.h. (1),  $\Gamma', \phi_2 \Vdash \phi_2$

By closing property  $\Gamma' \Vdash \phi_2$ .

Case  $\phi \equiv \phi_1 \rightarrow \phi_2$

$$(2) \quad \Gamma \Vdash \phi_1 \rightarrow \phi_2 \Rightarrow \Gamma \vdash_{cf} \phi_1 \rightarrow \phi_2$$

-----

By i.h. (1)  $\Gamma, \phi_1 \Vdash \phi_1$

By  $\bigcirc$ ,  $\Gamma, \phi_1 \Vdash \phi_2$

By i.h. (2)  $\Gamma, \phi_1 \vdash_{cf} \phi_2$

Hence  $\Gamma \vdash_{cf} \phi_1 \rightarrow \phi_2$  by  $(\rightarrow R)$ .

Case  $\phi \equiv \phi_1 \vee \phi_2$

(1)  $\phi_1 \vee \phi_2 \in \Gamma \Rightarrow \Gamma \Vdash \phi_1 \vee \phi_2$   
-----

We have  $\{(\Gamma, \phi_1), (\Gamma, \phi_2)\} \triangleright_{ctx} \Gamma$

By i.h. (1),  $\Gamma, \phi_1 \Vdash \phi_1$  and  $\Gamma, \phi_2 \Vdash \phi_2$ .

Hence  $\Gamma \vdash \phi_1 \vee \phi_2$ .

Case  $\phi \equiv \phi_1 \vee \phi_2$

$$(2) \quad \Gamma \Vdash \phi_1 \vee \phi_2 \Rightarrow \Gamma \vdash_{cf} \phi_1 \vee \phi_2$$

-----

By  $\circ$ , there is  $C \triangleright_{ctx} \Gamma$  such that

for every  $\Gamma' \in C$ ,  $\Gamma' \Vdash \phi_1$  or  $\Gamma' \Vdash \phi_2$ .

So for every  $\Gamma' \in C$ ,  $\Gamma' \vdash_{cf} \phi_1$  or  $\Gamma' \vdash_{cf} \phi_2$ , by i.h (2).

So for every  $\Gamma' \in C$ ,  $\Gamma' \vdash_{cf} \phi_1 \vee \phi_2$ , by ( $\vee R$ ) rules.

Since  $\frac{C}{\Gamma}$  is admissible,  $\Gamma \vdash_{cf} \phi_1 \vee \phi_2$ , by sequencing property.





# Sieve

In a coverage (rather than coverage base) we only allow covers  $C \triangleright w$ , where  $C$  is a sieve on  $w$ .

A set  $C \subseteq W$  is a sieve on  $w \in W$  if

- $C \subseteq \downarrow w$
- $C = \downarrow C$  ( $C$  is down closed)

If  $C$  is a sieve on  $w$  and  $v \leq w$  then

$$C \cdot v := \{u \in C \mid u \leq v\}$$

is a sieve on  $v$ .

$$w \mapsto \text{Sieves}(w)$$

defines a presheaf

$$W^{op} \rightarrow \underline{\text{Set}}$$

# Coverage

A coverage is a relation  $C \triangleright w$  between  $w \in W$  and sieves  $C$  on  $w$  satisfying

Reflexivity  $\downarrow w \triangleright w$

Transitivity If  $C \triangleright w$  and  $S$  is a sieve on  $w$  satisfying

$$\forall v \in C, S \cdot v \triangleright v$$

then  $S \triangleright w$ .

Stability If  $C \triangleright w$  and  $v \leq w$  then  $C \cdot v \triangleright v$ .

## Proposition (Properties of coverage)

Let  $\triangleright$  be a coverage. Then  $\triangleright$  satisfies

Monotonicity If  $C \triangleright w$  and  $S$  is a sieve on  $w$  then

$$C \subseteq S \Rightarrow S \triangleright w$$

Intersection If  $C \triangleright w$  and  $C' \triangleright w$  then  $C \cap C' \triangleright w$

Proof hints For monotonicity use reflexivity + transitivity.

For intersection use transitivity + stability.

## Coverages vs. Coverage bases

### Proposition

(1) If  $\triangleright$  is a coverage base then

$$C \triangleright w \quad :\Leftrightarrow \quad \exists B \subseteq C, \quad B \triangleright w \text{ and } C = \downarrow B$$

is a coverage. We say  $\triangleright$  is the coverage generated by  $\triangleright$ .

See  
Warning  
at end  
of part 2

(2) Every coverage  $\triangleright$  is generated by the coverage base

$$B \triangleright w \quad :\Leftrightarrow \quad \downarrow B \triangleright w.$$

Proof Exercise!

## The $\neg\neg$ -dense coverage

On any poset  $W$  define

$$C \triangleright_{\neg\neg} w \iff C \text{ is a sieve on } w \text{ and} \\ \forall v \leq w \neg\neg \exists u \leq v \ u \in C.$$

Proposition  $\triangleright_{\neg\neg}$  is a coverage.

Proof Exercise.

$\neg\neg$  Dense  $\Rightarrow$  Classical

Proposition Using the coverage  $\triangleright_{\neg\neg}$ ,  $\forall w \in W$  with  $\neg\neg\phi \rightarrow \phi$

Proof Suppose  $w \Vdash \neg\neg\phi$

we show  $\{u \leq w \mid u \Vdash \phi\} \triangleright_{\neg\neg} w$ . Then  $w \Vdash \phi$  by cover property.

$\{u \leq w \mid u \Vdash \phi\}$  is a sieve by monotonicity of  $\Vdash$

Suppose  $v \leq w$ . We need to show  $\neg\neg \exists u \leq v, u \Vdash \phi$ .

Suppose for  $\perp$  that  $\neg \exists u \leq v, u \Vdash \phi$ .

Then  $v \Vdash \neg\phi$ .

By  $\circ$ ,  $v \Vdash \perp$ . So  $\emptyset \triangleright_{\neg\neg} v$ . So  $\forall v' \leq v$   $\neg\neg \exists u' \leq v, u' \Vdash \phi$ . I.e.  $\perp$ .



# Sheaves & Proofs

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## Part 2

# Sheaf Semantics over Categories

(The logic of Grothendieck toposes)

- Presheaf semantics
- Coverage semantics over categories
- Sheaves for coverages
- Extending a separated model to a sheaf model



Let us attempt to directly generalise our Cut-free Completeness proof from propositional logic to intuitionistic predicate logic

We try to define appropriate  $\mathcal{W}_{\text{ctx}}$  and  $\triangleright_{\text{ctx}}$

$\mathcal{W}_{\text{ctx}} := \{ \Theta; \Gamma \mid \Theta \text{ a finite set of variables, } \Gamma \text{ a finite set of } \Theta\text{-formulas} \}$

The set of contexts

$$\Theta'; \Gamma' \leq \Theta; \Gamma \iff \Theta'; \Gamma' \supseteq \Theta; \Gamma$$

$$C \triangleright_{\text{ctx}} \Theta; \Gamma \iff \frac{C}{\Theta; \Gamma} \text{ is a derivable context rule}$$

# Context rules

$$\frac{\Theta; \Gamma, \phi, \phi'}{\Theta; \Gamma \ni \phi \wedge \phi'}$$

$$\frac{\Theta \oplus x; \Gamma, \phi}{\Theta; \Gamma \ni \exists x \phi}$$

$$\frac{\Theta; \Gamma', \phi \quad \Theta; \Gamma, \phi'}{\Theta; \Gamma \ni \phi \vee \phi'}$$

$$\frac{\Theta; \Gamma', \phi[t/x]}{\Theta; \Gamma \ni \forall x \phi}$$

$$\frac{}{\Theta; \Gamma \ni \perp}$$

$$\frac{\Theta; \Gamma, \phi'}{\Theta; \Gamma \ni \phi \rightarrow \phi'} \quad \Theta; \Gamma \vdash_{cf} \phi$$

However, the weakening property fails for derivable context rules.

E.g.  $\frac{y ; \exists x P(x), P(y)}{; \exists x P(x)}$  is a derivable context rule

but  $\frac{y ; \exists x P(x), P(y), Q(y)}{y ; \exists x P(x), Q(y)}$  is not a derivable context rule

Indeed  $\triangleright_{\text{ctx}}$  does not satisfy the stability property of a coverage base.

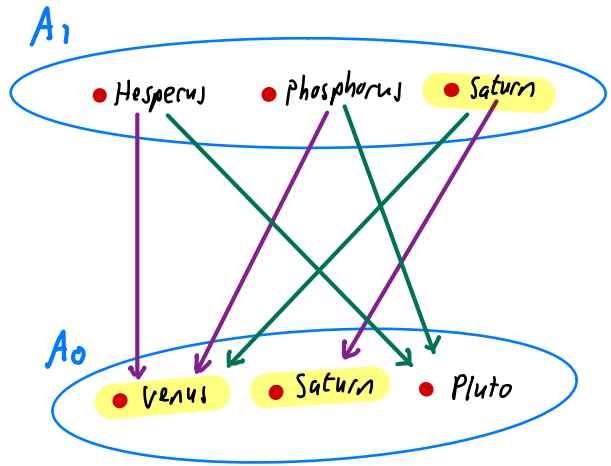
An elegant solution to this problem is to replace the partial order  $\mathbb{W}_{\text{ctx}}$  with a category.

# Presheaf semantics

A Presheaf Model is given by

- A (small) category  $\mathbb{C}$  of Worlds
- To every  $X \in \mathbb{C}$  a set  $A_X$
- Transition functions  $(t_f : A_X \rightarrow A_Y)_{Y \xrightarrow{f} X}$   
Satisfying  $Z \xrightarrow{g} Y \xrightarrow{f} X \Rightarrow t_{f \circ g} = t_g \circ t_f$
- For every predicate  $P$  of arity  $k$   
Subsets  $P_w \subseteq (A_w)^k$  satisfying  
 $Y \xrightarrow{f} X \Rightarrow t_f(P_X) \subseteq P_Y$

$$\mathbb{C} = \begin{matrix} & & 1 \\ & \nearrow f_0 & \nearrow f_1 \\ 0 & & \end{matrix}$$



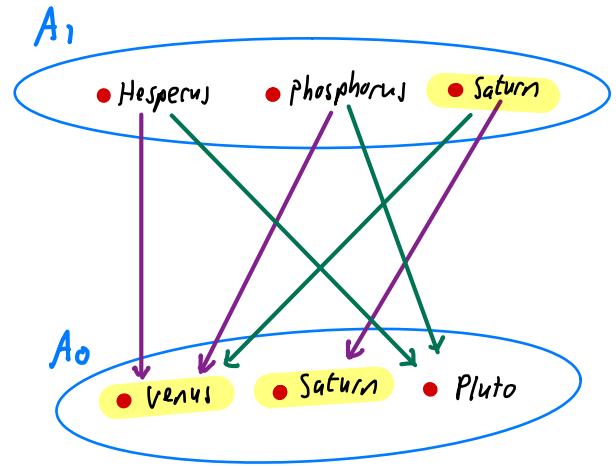
 The planet predicate

# Category-theoretic Formulation

A Presheaf model is given by

- A small category  $\mathcal{C}$  of worlds
- A functor  $A: \mathcal{C}^{op} \rightarrow \underline{\text{Set}}$   
(i.e., a presheaf  $A \in \text{Psh}(\mathcal{C})$ )
- For every predicate  $P$  of arity  $k$   
a subpresheaf  $R \subseteq A^k$

$$\mathcal{C} = \begin{matrix} & & 1 \\ & \nearrow f_0 & \searrow f_1 \\ 0 & & 0 \end{matrix}$$



 The planet predicate

## Notation

$\text{Venus} = t_{f_0}(\text{Hesperus})$  (previous slide)  
 $= A(f_0)(\text{Hesperus})$  (this slide)  
 $= \text{Hesperus} \cdot f_0$  (henceforth)

Presheaf forcing relation

$$X \Vdash \phi$$

$$X \Vdash P(a_1, \dots, a_k) \iff (a_1, \dots, a_k) \in P_x$$

$$X \Vdash a_1 = a_2 \iff a_1 = a_2$$

$$X \Vdash \phi \wedge \psi \iff X \Vdash \phi \text{ and } X \Vdash \psi$$

$$X \Vdash \phi \vee \psi \iff X \Vdash \phi \text{ or } X \Vdash \psi$$

$$X \Vdash \perp \iff \text{false}$$

$$X \Vdash \exists x \phi \iff \text{there exists } a \in A_x \text{ s.t. } X \Vdash \phi[x := a]$$

$$X \Vdash \phi \rightarrow \psi \iff \text{for all } Y \xrightarrow{f} X \quad Y \Vdash \phi \cdot f \text{ implies } Y \Vdash \psi \cdot f$$

$$X \Vdash \forall x \phi \iff \text{for all } Y \xrightarrow{f} X \text{ and } a \in A_Y \quad Y \Vdash (\phi \cdot f)[x := a]$$

## Meta theorems for Presheaf Semantics

Monotonicity If  $X \Vdash \phi$  and  $Y \xrightarrow{f} X$  then  $Y \Vdash \phi$ .

Soundness If  $\phi$  is provable in intuitionistic predicate logic  
then for all **presheaf models**  $(\mathbb{C}, \dots)$  and  $x \in \mathbb{C}$ ,  $x \Vdash \phi$ .

Completeness If, for all **presheaf models**  $(\mathbb{C}, \dots)$  and  $x \in \mathbb{C}$ ,  $x \Vdash \phi$   
then  $\phi$  is provable in intuitionistic predicate logic

(Completeness requires a classical meta-theory.)

# Coverage base

a.u.a. base for a  
Grothendieck topology

on a category

A coverage base on a category  $\mathcal{C}$  is a relation of the form  $C \triangleright X$ ,

where  $X \in \mathcal{C}$  and  $C \subseteq \bigcup_{Y \in \mathcal{C}} \mathcal{C}(Y, X)$

Reflexivity

$$\{x \xrightarrow{\text{id}_x} x\} \triangleright x$$

Transitivity

If  $C \triangleright X$  and, for all  $Y \xrightarrow{f} X \in C$ ,  $C_f \triangleright Y$

then  $\{z \xrightarrow{g \circ f} x \mid Y \xrightarrow{f} X \in C, z \xrightarrow{g} Y \in C_f\} \triangleright x$

Stability

If  $C \triangleright x$  and  $x' \xrightarrow{f'} x$  then there exists  $D \triangleright x'$  such that,  
for all  $y' \xrightarrow{g'} x' \in D$  there exist  $y \xrightarrow{g} x \in C$  and  $y' \xrightarrow{f'} x'$

such that

$$\begin{array}{ccc} y' & \xrightarrow{f'} & y \\ g' \downarrow & & \downarrow g \\ x' & \xrightarrow{f} & x \end{array}$$



A separated model is given by

- A small category  $\mathbb{C}$  of worlds with coverage base  $\triangleright$
- To every  $x \in \mathbb{C}$  a set  $A_x$
- Transition functions  $(\mapsto \cdot f : A_x \rightarrow A_y)_{y \xrightarrow{f} x}$  s.t.  $(a \cdot f) \cdot g = a \cdot (f \circ g)$
- $\mathbb{C} \triangleright X \Rightarrow \forall a, b \in A_x (\forall y \xrightarrow{f} x \in \mathbb{C}, a \cdot f = b \cdot f) \Rightarrow a = b$  . ( $\triangleleft$ -separated)
- For every predicate  $P$  of arity  $k$  subsets  $P_w \subseteq (A_w)^k$  satisfying
  - $\forall \vec{a} \in A_x^k (\vec{a} \in P_x \Rightarrow \forall y \xrightarrow{f} x, \vec{a} \cdot f \in P_y)$
  - $\mathbb{C} \triangleright X \Rightarrow \forall \vec{a} \in A_x^k (\forall y \xrightarrow{f} x \in \mathbb{C}, \vec{a} \cdot f \in P_y) \Rightarrow \vec{a} \in P_x$  ( $P$  is  $\triangleleft$ -closed)

## Category-theoretic Formulation

A separated Model is given by

- A small category  $\mathcal{C}$  of Worlds with coverage base  $\triangleright$
- A  $\triangleright$ -separated presheaf  $A : \mathcal{C}^{op} \rightarrow \underline{\text{Set}}$
- For every predicate  $P$  of arity  $k$ , a  $\triangleright$ -closed subpresheaf  $P \subseteq A^k$

## Forcing w.r.t. a coverage

$$X \Vdash \phi \vee \psi \Leftrightarrow$$

there exists a cover  $C \triangleright X$  such that,

for every  $Y \xrightarrow{f} X \in C$ ,  $Y \Vdash \phi \cdot f$  or  $Y \Vdash \psi \cdot f$

$$X \Vdash \perp \Leftrightarrow \emptyset \triangleright X$$

$X \Vdash \exists x \phi \Leftrightarrow$  there exists a cover  $C \triangleright X$  such that,

for every  $Y \xrightarrow{f} X \in C$

there exists  $a \in \underline{A}_Y$  s.t.  $Y \Vdash (\phi \cdot f)[x := a]$

## Meta theorems for sheaf semantics

Monotonicity If  $X \Vdash \phi$  and  $Y \xrightarrow{f} X$  then  $Y \Vdash \phi \cdot f$

Cover property If  $C \triangleright X$  satisfies, for all  $Y \xrightarrow{f} X \in C$ ,  $Y \Vdash \phi \cdot f$  then  $X \Vdash \phi$ . ] See warning at end of part 2

Soundness If  $\phi$  is provable in intuitionistic predicate logic then for all *separated models*  $(C, \triangleright, \dots)$  and  $X \in C$ ,  $X \Vdash \phi$ .

Completeness If, for all *separated models*  $(C, \triangleright, \dots)$  and  $X \in C$ ,  $X \Vdash \phi$  then  $\phi$  is provable in intuitionistic predicate logic

Completeness has an elegant proof in a constructive meta-theory

To prove completeness we use a category based model

The category  $\mathbb{C}_{ctx}$

Objects:  $\Theta; \Gamma$  (left-hand sides of sequents)

Morphisms:  $\sigma: \Theta'; \Gamma' \rightarrow \Theta; \Gamma$

$\sigma: \Theta \rightarrow \Theta'$  is an injective function

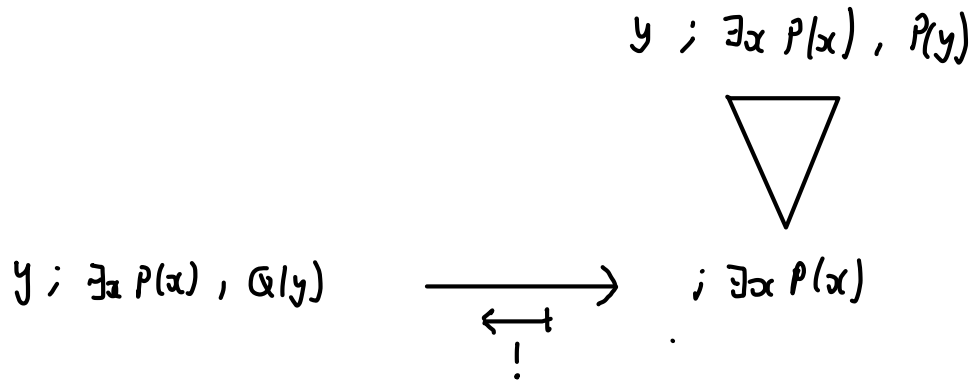
and  $\Gamma[\sigma] \subseteq \Gamma'$

The coverage base  $\triangleright_{ctx}$

$C \triangleright \Theta; \Gamma \iff \frac{C}{\Theta; \Gamma}$  is a derivable context rule.

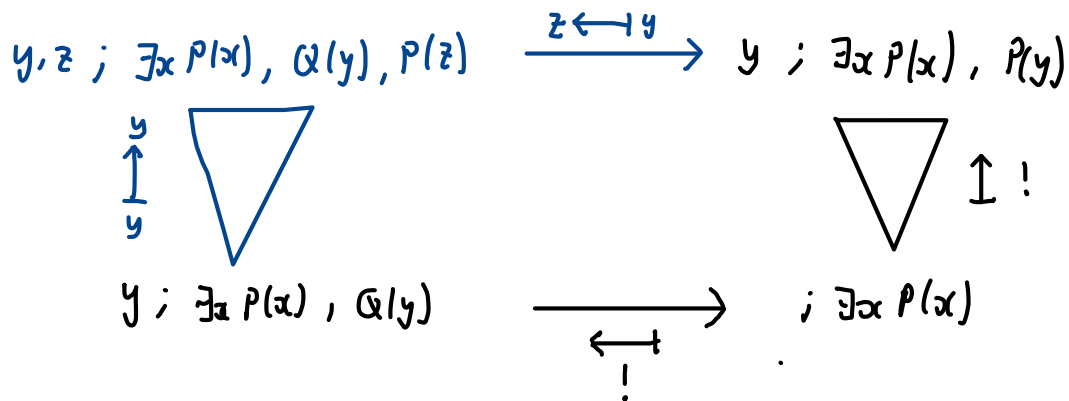
$\triangleright_{ctx}$  is a coverage base

Example illustrating stability



$\Delta_{\text{ctx}}$  is a coverage base

Example illustrating stability



Definition of full separated model  $\mathcal{M}_{ctx}$

- Category of worlds  $\mathcal{C}_{ctx}$
- Coverage base  $\Delta_{ctx}$
- Separated presheaf

$$A_{\Theta; \Gamma} := \{t \mid \text{vars}(t) \subseteq \Theta\} / \sim_{\Theta; \Gamma}$$

$$t \sim_{\Theta; \Gamma} t' :\Leftrightarrow \forall \Theta'; \Gamma' \xrightarrow{\sigma} \Theta; \Gamma \text{ and } \Theta' \vdash [z] \text{-formula } \Psi[z]$$

$$\Theta'; \Gamma' \vdash_{cf} \Psi[t[\sigma]] \Leftrightarrow \Theta'; \Gamma' \vdash_{cf} \Psi[t'[\sigma]]$$

$$[t] \cdot \sigma := [t[\sigma]]$$

- Interpretation of predicates

$$([t_1], \dots, [t_n]) \in P_{\Theta; \Gamma} :\Leftrightarrow \Theta; \Gamma \vdash_{cf} P(t_1, \dots, t_n)$$



The proof of cut-free completeness

entailment w.r.t.

category-based separated models.

$$\Theta; \Gamma \models \psi \Rightarrow \Theta; \Gamma \vdash_{cf} \psi$$

now proceeds along similar lines to the propositional case

Once again, the main step is to prove

Main lemma  $\mathcal{N}_{ctx}$  enjoys the following.

For any  $\Theta; \Gamma \in \mathbb{C}_{ctx}$  and  $\Theta$ -formula  $\phi$

(1) if  $\phi \in \Gamma$  then  $\Theta; \Gamma \Vdash \phi[\vec{x} := \overrightarrow{[x]}]$

(2) if  $\Theta; \Gamma \Vdash \phi[\vec{x} := \overrightarrow{[x]}]$  then  $\Theta; \Gamma \vdash_{cf} \phi$ .

We have obtained completeness for intuitionistic predicate logic with respect to category-based separated models.

$$(\forall \mathcal{N} = (\mathcal{C}, \Delta, \dots), \theta; \Gamma \models_{\mathcal{N}} \psi) \Rightarrow \theta; \Gamma \vdash \psi$$

This can be strengthened to completeness with respect to poset-based separated models

$$(\forall \mathcal{M} = (\mathcal{W}, \Delta, \dots), \theta; \Gamma \models_{\mathcal{M}} \psi) \Rightarrow \theta; \Gamma \vdash \psi$$

using the **Diaconescu cover**, which can be used to construct a poset-based model out of a category-based model, preserving logical properties.

# Sheaf models

## Separated Model

- A small category  $\mathcal{C}$  of worlds with coverage base  $\triangleright$
- A Separated presheaf  $A : \mathcal{C}^{op} \rightarrow \underline{\text{Set}}$
- For every predicate  $P$  of arity  $k$ , a closed subpresheaf  $P \subseteq A^k$

## Sheaf model (a special case of separated model)

- A small category  $\mathcal{C}$  of worlds with coverage base  $\triangleright$
- A sheaf  $A : \mathcal{C}^{op} \rightarrow \underline{\text{Set}}$  (every sheaf is a fortiori separated)
- For every predicate  $P$  of arity  $k$ , a closed subpresheaf  $P \subseteq A^k$   $\equiv$  subsheaf

## Sheaf on a topological space $T$

Given a presheaf  $A: \mathcal{O}(T)^{op} \rightarrow \underline{\text{Set}}$

- A matching family for a cover  $C \triangleright w$  is a family  $(a_v \in A_v)_{v \in C}$  such that  $\forall u, v \in C \quad a_u \cdot (u \cap v) = a_v \cdot (u \cap v)$
- An amalgamation for a matching family  $(a_v \in A_v)_{v \in C}$  is an element  $a_w \in A_w$  s.t.  $\forall v \in C \quad a_v = a_w \cdot v$ .
- The presheaf  $A$  is a sheaf if every matching family has a unique amalgamation. (Separated  $\Leftrightarrow$  at most one amalgamation!)

Example sheaf  $C: \mathcal{O}(T) \rightarrow \underline{\text{Set}} \quad C(u) := \{f: u \rightarrow \mathbb{R} \mid f \text{ continuous}\}$   
 $f \cdot v := f|_v \quad \text{for } f \in C(u) \text{ and } v \subseteq u$

## Sheaf for a coverage base (over a poset)

Given a presheaf  $A: W^{op} \rightarrow \underline{\text{Set}}$  and coverage base  $\triangleright$  on  $W$ .

- A matching family for a cover  $C \triangleright w$  is a family  $(a_v \in A_v)_{v \in C}$  such that  $\forall v, v' \in C \quad \forall u \leq v, v', \quad a_v \cdot u = a_{v'} \cdot u$
- An amalgamation for a matching family  $(a_v \in A_v)_{v \in C}$  is an element  $a_w \in A_w$  s.t.  $\forall v \in C \quad a_v = a_w \cdot v$ .
- The presheaf  $A$  is a sheaf if every matching family has a unique amalgamation. (Separated  $\Leftrightarrow$  at most one amalgamation!)

# Sheaf for a coverage base on a category

Given a presheaf  $A: \mathcal{C}^{op} \rightarrow \underline{\text{Set}}$  and coverage  $\triangleright$  on  $\mathcal{C}$ .

- A matching family for a cover  $C \triangleright X$  is a family  $(a_f \in A_Y)_{Y \xrightarrow{f} X \in C}$  such that  $\forall Y \xrightarrow{f} X, Y' \xrightarrow{f'} X \in C, \forall Z \xrightarrow{g} Y$  in  $\mathcal{C}$ ,  $a_f \cdot g = a_{f'} \cdot g'$ .  

$$\begin{array}{ccc} & g & \\ & \downarrow & \\ g' & \downarrow & \\ Y' & \xrightarrow{f'} & X \end{array}$$
- An amalgamation for a matching family  $(a_f \in A_Y)_{Y \xrightarrow{f} X \in C}$  is an element  $a \in A_X$  s.t.  $\forall Y \xrightarrow{f} X \in C, a_f = a \cdot f$ .
- The presheaf  $A$  is a sheaf if every matching family has a unique amalgamation. (Separated  $\Leftrightarrow$  at most one amalgamation!)

There is no loss of generality in considering sheaf models

### Theorem

Let  $(\mathbb{C}, \triangleright)$  be a site (small category + coverage (base)).

Every separated model  $\mathcal{N}$  over  $(\mathbb{C}, \triangleright)$  has a dense elementary embedding into a sheaf model  $\mathcal{N}^+$  over  $(\mathbb{C}, \triangleright)$ .

$$\mathcal{N} \hookrightarrow \mathcal{N}^+$$

Moreover, this characterises  $\mathcal{N}^+$  up to isomorphism.

Given presheaves  $A, B : \mathbb{C}^{op} \rightarrow \underline{\text{Set}}$

A presheaf map  $\alpha : A \rightarrow B$  is a family of functions

$$(\alpha_x : A_x \rightarrow B_x)_{x \in \mathbb{C}}$$

such that  $\forall y \xrightarrow{f} x$  in  $\mathbb{C}$  and  $a \in A_x$ ,  $\alpha_y(a \cdot f) = (\alpha_x(a)) \cdot f$ .

I.e., a presheaf map is a *natural transformation*  $\alpha : A \Rightarrow B$ .

$\alpha$  is an embedding of presheaves if every  $\alpha_x$  is injective.

$\alpha$  is dense w.r.t.  $\triangleright$  if

$$\forall b \in B_x \exists C \triangleright x \text{ s.t. } \forall Y \xrightarrow{f} X \in \mathbb{C}, \exists a \in A_Y \text{ s.t. } \alpha_Y(b) = a \cdot f.$$



Given separated models  $\mathcal{N} = (A, (P)_P)$  and  $\mathcal{N}' = (A', (P')_P)$

an embedding of models is a presheaf embedding  $\alpha: A \rightarrow A'$  that further satisfies

For all  $x \in \mathbb{C}$  and  $a_1, \dots, a_n \in A_x$

$$(a_1, \dots, a_n) \in P_x \iff (\alpha_x(a_1), \dots, \alpha_x(a_n)) \in P'_x$$

It is an elementary embedding if, for all formulas  $\phi$ ,

$$x \Vdash^{\mathcal{N}} \phi(a_1, \dots, a_n) \iff x \Vdash^{\mathcal{N}'} \phi(\alpha_x(a_1), \dots, \alpha_x(a_n))$$

# Grothendieck's $(\cdot)^+$ construction

Let  $\mathbb{C}$  be a small category with coverage base  $\triangleright$ .

Given a presheaf  $A: \mathbb{C} \rightarrow \underline{\text{Set}}$  define  $A^+ : \mathbb{C} \rightarrow \underline{\text{Set}}$

$$A_x^+ := \left\{ (c, (a_f)_{f \in c}) \mid c \triangleright x, (a_f) \text{ is a matching family for } c \right\} / \sim$$

$$(c, (a_f)_f) \sim (c', (a'_f)_{f'}) \iff \text{for every commuting square in } \mathbb{C}$$

$$\begin{array}{ccc} \bullet & \xrightarrow{g} & \bullet \\ g' \downarrow & & \downarrow f \in c \\ \bullet & \xrightarrow{f' \in c'} & x \end{array} \quad , \quad a_f \cdot g = a'_{f'} \cdot g'$$

## Theorem (Grothendieck)

- $A^+$  is separated, for any presheaf  $A$ .
- If  $A$  is separated, then  $A^+$  is a sheaf,  
and  $a \in A_x \mapsto [(\{\text{id}_x\}, \{a\})] : A \rightarrow A^+$   
is a dense embedding of presheaves.

## Extending a model

### Theorem

Suppose  $\alpha: A \Rightarrow A'$  is a dense embedding between separated presheaves.

Let  $\mathcal{N} = (A, (P)_P)$  be any separated model.

For any  $k$ -ary predicate  $P$ , define  $P' \subseteq (A')^k$  by

$$(a'_1, \dots, a'_k) \in P'_x : \Leftrightarrow \exists C \supset X \text{ s.t. } \forall y \xrightarrow{f} x \in C$$

$$\exists a_1, \dots, a_k \in A_y . \alpha_y(a_1) = a'_1 \cdot f , \dots , \alpha_y(a_k) = a'_k \cdot f \\ \text{and } (a_1, \dots, a_k) \in P_y$$

Then  $\mathcal{N}' := (A', (P')_P)$  is a separated model and  $\alpha: \mathcal{N} \Rightarrow \mathcal{N}'$  is an elementary embedding.

## Proof

We just look at the elementary embedding property.

We prove

$$X \models^N \phi(a_1, \dots, a_n) \iff X \models^{N'} \phi(\alpha_X(a_1), \dots, \alpha_X(a_n))$$

by induction on  $\phi$ .

We consider just one case.

Case  $\phi \equiv \exists x \phi'$

$\Rightarrow$  Suppose  $X \Vdash (\exists x \phi')[a_1, \dots, a_n]$

Then exists  $C \triangleright X$  s.t.  $\forall Y \xrightarrow{F} X \in C, \exists a \in X_Y$  s.t.

$$Y \Vdash^N \phi'[a_1.f, \dots, a_n.f, a]$$

By i.h.,  $Y \Vdash^{N'} \phi'[\alpha_Y(a_1.f), \dots, \alpha_Y(a_n.f), \alpha_Y(a)]$

i.e.,  $Y \Vdash^{N'} \phi'[\alpha_X(a_1).f, \dots, \alpha_X(a_n).f, \alpha_Y(a)]$

Hence  $X \Vdash^{N'} (\exists x \phi')[\alpha_X(a_1), \dots, \alpha_X(a_n)]$ .

$\Leftarrow$

Suppose  $X \Vdash^{N'} (\exists x \phi') [\alpha_x(a_1), \dots, \alpha_x(a_n)]$

There exists  $C \triangleright X$  s.t.  $\forall Y_f \xrightarrow{f} X, \exists a'_f \in A'_{Y_f}$  s.t.

$$Y_f \Vdash^{N'} \phi' [\alpha_x(a_1) \cdot f, \dots, \alpha_x(a_n) \cdot f, a'_f]$$

By density  $\exists C_f \triangleright Y_f \forall z \xrightarrow{g} Y_f \in C_f \exists a_g \in A_z \alpha_z(a_g) = a'_f \cdot g$ .

By transitivity of  $\triangleright$ ,  $D := \{f \circ g \mid f \in C, g \in C_f\} \triangleright X$  warning!

Moreover,  $\forall z \xrightarrow{f \circ g} X \in D, z \Vdash^{N'} \phi' [\alpha_z(a_1 \cdot f \cdot g), \dots, \alpha_z(a_n \cdot f \cdot g), \alpha_z(a_g)]$ .

By i.h.,  $z \Vdash^N \phi' [a_1 \cdot f \cdot g, \dots, a_n \cdot f \cdot g, a_g]$

Hence,  $z \Vdash^N (\exists x \phi') [a_1, \dots, a_n]$ .

□

The **warning** is that  $\{f \circ g \mid f \in C, g \in C_f\}$  is not a good definition, as it uses choice to select a chosen  $C_f$  for every  $f$ .

(Similar issues arise in proving that  $\Pi$  enjoys the cover property in the case of  $\vee$  and  $\exists$ , and in proving transitivity for the coverage generated by a coverage base. With the given derivation of coverage base, these proofs use choice.)

Two possible solutions

- Define  $\Pi$  w.r.t. a coverage rather than coverage base.
- Or work with coverage bases formulated using indexed families

$$\{Y_i \xrightarrow{f_i} X\}_{i \in I} \triangleright X.$$



# Coverage bases using indexed families

A coverage base is a relation  $\{Y_i \xrightarrow{f_i} X\}_{i \in I} \triangleright X$  between objects  $X$  in  $\mathcal{C}$  and indexed families of maps into  $X$  satisfying

- (Reflexivity)  $\{X \xrightarrow{id} X\}_{i \in 1} \triangleright X$
- (Transitivity)  $\{Y_i \xrightarrow{f_i} X\}_{i \in I} \triangleright X$  and  $\forall i \{Z_{ij} \xrightarrow{g_{ij}} Y_i\}_{j \in J_i} \triangleright Y_i \Rightarrow \{Z_{ij} \xrightarrow{f_i \circ g_{ij}} X\}_{(i,j) \in \sum_{i \in I} J_i} \triangleright X$ .
- (Stability)  $\{Y_i \xrightarrow{f_i} X\}_{i \in I} \triangleright X$  and  $X' \xrightarrow{g} X \Rightarrow \exists \{Y'_j \xrightarrow{f'_{ij}} X'\}_{j \in J} \triangleright X'$   
s.t.  $\forall j \in J \exists i \in I \exists Y'_j \xrightarrow{g'} Y_i$  s.t.  $g \circ f'_{ij} = f_i \circ g'$ .
- (Reindexing)  $\{Y_i \xrightarrow{f_i} X\}_{i \in I}$  and  $r: J \rightarrow I$  surjective  $\Rightarrow \{Y_{s(i)} \xrightarrow{f_{s(i)}} X\}_{i \in J} \triangleright X$ .

# Perspective

Coverage  $\equiv$  Grothendieck topology

Coverage base  $\equiv$  base for a Groth. top. (one sees other variant definitions)

The category of presheaf maps between sheaves is a Groth. topos.

All Groth. toposes arise in this way.

A sheaf model  $\mathcal{N}$  over  $(\mathbb{C}, \triangleright)$  is just

- An object  $A \in \mathcal{E} := \underline{\text{Sh}}(\mathbb{C}, \triangleright)$
- For every predicate  $P$  a subobject  $P \twoheadrightarrow A^k$  in  $\mathcal{E}$

That is, it is an ordinary relational structure from the perspective of  $\mathcal{E}$ .

The forcing relation captures truth in the internal logic of  $\mathcal{E}$ . E.g.,

$$\forall x \in \mathbb{C}, x \Vdash \phi \quad \Leftrightarrow \quad \mathcal{E} \text{ validates } \mathcal{N} \models \phi$$

Internal  
Tarskian  
semantics

## Literature (will be expanded in due course.)

The literature I know on sheaf semantics approaches it via first understanding toposes and their logic, requiring substantial category theory.

My favourite book that takes this approach is

- Mac Lane & Moerdijk *Sheaves in Geometry and Logic*

A more gently paced presentation (with less content) can be found in

- Goldblatt *Topoi*

# Proofs of completeness in intuitionistic meta-theories

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- W. Veldman *An intuitionistic completeness theorem ...* 1976
- H. De Swart *Another intuitionistic completeness proof* 1976

Completeness addressed by Brouwerian intuitionists, using the Fan theorem.  
The Veldman paper introduces "exploding worlds" in which  $\perp$  holds

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- T. Coquand & J. Smith *An application of constructive completeness* 1996
- E. Palmgren 1997 *Constructive sheaf semantics* 1997

Papers that use coverage-based models to achieve constructive completeness.

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- W. Buchholz *Ein ausgezeichnetes Modell für die intuitionistische Typenlogik* 1975
- J. Avigad *Algebraic proofs of cut elimination* 2000

The coverage-based proof of cut-elimination presented in these lectures.